



AUCKLAND SITE · AOTEAROA NEW ZEALAND · MON 19 – TUE 20 OCTOBER 2026

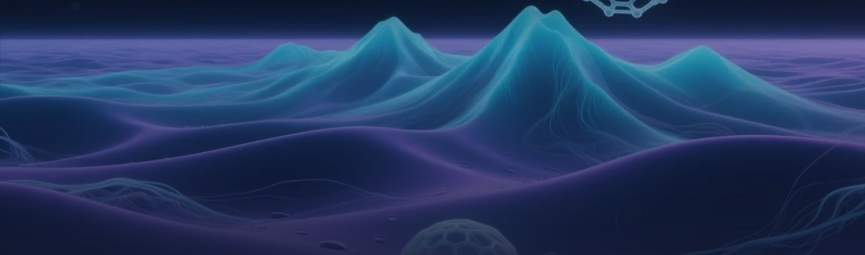
How far can a chain of **surrogates** take us in biomolecular interactions?

Biology has a ladder of scales and machine learning now has one alongside it, most of it open-weight. **Two days finding out how far up you have to climb before the answer can be trusted.**

SYSTEMS · PROTEINS · LIGAND BINDING · DRUG DISCOVERY · PHARMACOLOGY

THEME 1 MANY MOLECULES, COARSE

Screening at scale

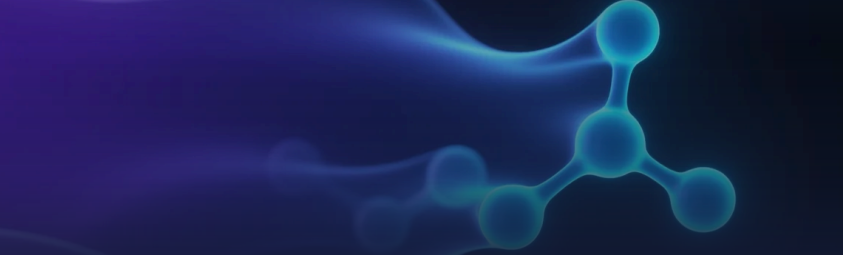


Models that know something about shape, used to filter before you pay for docking.

Boltz-2 · OpenFold3 · IntFold · ChemBERTa-2 · ESM-2

THEME 2 ATOMS, IN MOTION

Molecules in motion

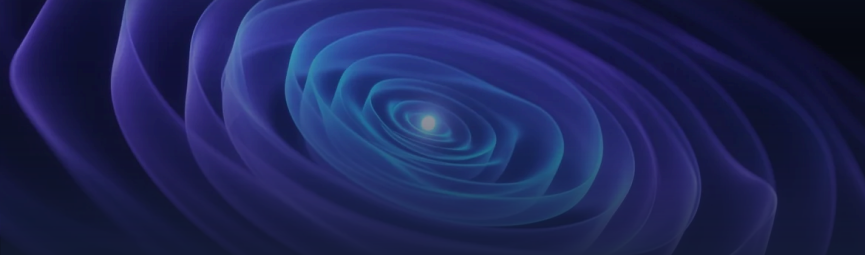


Machine-learned interatomic potentials in MD. Near-quantum forces you can afford to run.

MACE · UMA · eSEN · Orb-v3 · NequIP

THEME 3 ONE EVENT, WHOLE CELL

Binding to whole system

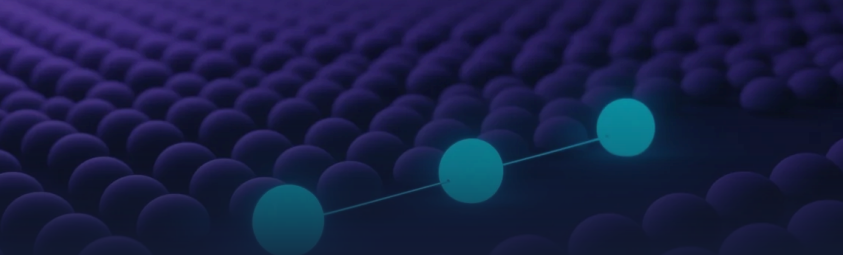


Carrying a molecular signal up to functional and physiological change.

STATE · scGPT · Geneformer · scFoundation

THEME 4 EVERY APPROVED DRUG

Repurposing what we have



Genotype, structure and approved drugs, with deliberately small models.

Evo 2 · Nucleotide Transformer · TxGNN · PrimeKG

WHO IT'S FOR

Bring your team, your expertise, and a problem you want to look at, or take one of our default targets. Chemists, physicists, structural biologists, statisticians, systems biologists, pharmacologists, clinicians, bioinformaticians, RSEs, and anyone new to it.

You don't need to be able to code. The agent writes the code and you bring the judgement about whether the answer is any good. If you can only make the opening, come for the opening and leave your team to it.

No stupid questions · Safe to experiment · A negative is a result

PRACTICAL

When	Mon 19 – Tue 20 October 2026
Where	Digital Research Innovation & AI Lab · Level 10, 70 Symonds Street
Cost	Free
Bring	A laptop
Day 1	Get running, report back
Day 2	Measure it, show what you got
Free All welcome Cross-disciplinary Food both days	

drai-inn.github.io/osi-hackathon-auckland

Nick Jones · njon001@aucklanduni.ac.nz · The global event runs 21–22 October and registration is open to anyone